Optimizing Management of Chemotherapy Induced Nausea and Vomiting (CINV)

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Objectives

- Define the types of chemotherapy-induced nausea and vomiting (CINV) (acute, delayed, anticipatory, breakthrough and refractory)
- Identify risk factors for CINV
- Classify the emetogenic potential of common chemotherapeutic agents
- Review CINV prophylaxis and management strategies.
- Discuss non-pharmacologic interventions and highlight the role of pharmacist in managing CINV.

The Clinical Impact of CINV

Chemotherapy-induced nausea and vomiting (CINV) is a clinical challenge that significantly impacts the quality of life, treatment adherence and can lead to complications such as dehydration and metabolic imbalances.

Consequences of Poor Control

- Dehydration, electrolyte imbalance, and nutritional deficiencies.
- Reduced quality of life and functional status
- Risk of non-adherence to treatment or dose reduction, compromising clinical outcome.

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National Comprehensive Cancer Network. NCCN Clinical Practice Guidelines in Oncology (NCCN Guidelines®) for Antiemesis. Version 2.2025. Published May 12, 2025. Accessed November 2, 2025. https://www.nccn.org/professionals/physician_gls/pdf/antiemesis.pdf
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Classification of CINV: Acute and Delayed

- Acute CINV typically occurs within 24 hours of chemotherapy administration.
- It is primarily mediated by serotonin release from enterochromaffin cells in the gastrointestinal tract.
- Consequently activating 5-HT3 receptors on vagal afferents and triggering the emetic reflex via the brainstem.

- Delayed CINV onset occurs more than 24 hours after chemotherapy administration.
- It typically peaks around 48 to 72 hours and can persist beyond 120 hours.
- It is primarily driven by substance P activation of the central neurokinin-1 receptors.

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Types of CINV: Anticipatory, Breakthrough and Refractory

- Anticipatory CINV occurs prior to treatment and develops as a conditioned response.
- It typically arises before chemotherapy due to previous negative experience in previous cycles.
- It is mediated by the psychological and limbic system rather than direct neurotransmitter release.

- Breakthrough CINV occurs at any time after chemotherapy, despite the use of adequate prophylaxis.
- It requires rescue antiemetics.

- Refractory CINV occurs when prophylaxis and/or rescue treatment is not effective.

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Patient Risk Factors for CINV

- Younger age
- Female sex
- Anxiety
- History of motion sickness or morning sickness.
- Low alcohol consumption
- Prior history of CINV

Treatment Related Risk Factors for CINV

- Emetogenic potential of the chemotherapy regimen
- Dose, schedule of chemotherapy
- Combination regimens
- Multi-day and oral regimens

Emetogenic Potential of Chemotherapeutic Agents

Level	Emetogenic Potential (% of Patients with Emesis)
1	Minimal (0 to <10%) e.g vinblastine, vincristine, rituximab, busulfan.
2	Low (10 to 30%) e.g trastuzumab, paclitaxel, fluorouracil, methotrexate, pemetrexed, gemcitabine.
3	Moderate (>30 to 90%) e.g carboplatin, cyclophosphamide dosed at < 1.5 g/m^2, doxorubicin, ifosfamide.
4	High (>90%) e.g cisplatin, cyclophosphamide dosed at 1.5 g/m^2, carmustine.

Information on the emetogenicity of chemotherapy agents is available from multiple sources, including the ASCO guidelines, NCCN, and drug references such as Lexicomp.

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Types of Antiemetics for CINV

Neurokinin-1 Receptor Antagonists (NK1 RAs)	Serotonin Receptor Antagonists (5-HT3 RAs)	Dopamine Receptor Antagonists	Others
MOA: inhibits the substance P/neurokinin-1 receptor, augmenting the antiemetic activity of 5-HT3 receptor antagonists and corticosteroid. **Examples**: aprepitant, fosaprepitant,	**MOA**: block serotonin, both peripherally on vagal nerve terminal and centrally in the chemoreceptor trigger zone. **Examples**: ondansetron, palonosetron	**MOA**: block dopamine receptors in the CNS, including in the chemoreceptor trigger zone which reduces signal to the vomiting center. Examples: olanzapine, prochlorperazine, promethazine, metoclopramide, haloperidol	**Corticosteroid**: MOA unknown. e.g dexamethasone **Benzodiazepine**: enhances GABA to decrease excitability and alleviates anxiety and anticipatory nausea and vomiting. e.g lorazepam 0.5-1 mg **Cannabinoid**: activates cannabinoid receptors within the CNS and inhibits the vomiting control mechanism in the medulla oblongata. e.g dronabinol, marinol **Anticholinergic** : inhibits the action of acetylcholine at the central and peripheral muscarinic receptors. e.g scopolamine patch

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Types of Antiemetics for CINV

Neurokinin-1 Receptor Antagonists (NK1 RAs)	Serotonin Receptor Antagonists (5-HT3 RAs)	Dopamine Receptor Antagonists	Others
Contraindications: rolapitant (do not use with thioridazine or pimozide) **Side effects:** abdominal pain, dizziness, dyspepsia, fatigue, hiccups and infusion reactions. **Clinical pearls:** aprepitant, fosaprepitant, netupitant can increase dexamethasone levels through CYP 3A4 inhibition. **Example**: aprepitant 125 mg po, day 1 and 80 mg on days 2-3. IV aprepitant 130 mg ~ 30 min prior to chemotherapy on day 1. fosaprepitant IV 150 mg on day 1 only ~ 30 min prior to chemotherapy.	**Contraindications**: do not use with apomorphine due to severe hypotension. **Side effects:** headache, constipation, fatigue **Warning:** dose -dependent QT prolongation (more common with IV), limit single dose of ondansetron to 16 mg; **the lowest risk is with palonosetron.** Risk of serotonin syndrome with other serotonergic drugs. **Example**: ondansetron oral: 8 mg po bid or 24 mg po daily. IV : 8 mg, max dose of 16 mg). Palonosetron 0.25 mg IV as a single dose ~ 30 mins prior to chemotherapy. Oral: 0.5 mg ~ 1 hr prior to chemotherapy.	**BW:** olanzapine, prochlorperazine, haloperidol can increase mortality in elderly with dementia related psychosis. Metoclopramide can cause tardive dyskinesia. **Warning**: avoid in parkinson disease **Side effects**: sedation, lethargy, acute extrapyramidal symptoms (antidote is benadryl or atropine), QT prolongation. **Example**: olanzapine 5-10 mg po daily	**Corticosteroid**: do not use in systemic fungal infections. **Side effects:** weight gain, fluid retention, insomnia, hyperglycemia. **Example**: dexamethasone 12 mg po /IV day 1, 8 mg days 2-4 **Cannabinoid**: caution with history of substance abuse. **Side effects:** somnolence, euphoria, increase appetite. **Anticholinergic** : **Side effects:** dry mouth, drowsiness, urinary retention.

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Antiemetic Regimens for Acute/Delayed CINV

INTRAVENOUS CHEMOTHERAPY RISK	ANTIEMETIC REGIMEN (DAY 1)	ANTIEMETIC REGIMEN (DAY 2 - 4 for Delayed CINV)
High emetic risk: 90 % frequency of acute emesis	3 or 4 drugs regimen ● Preferred: NK1 RA + 5-HT3 RA + olanzapine + dexamethasone ● Palonosetron + olanzapine + dexamethasone ● Avoid dexamethasone in cellular therapy	● Olanzapine continued at 2.5 - 10 mg qhs for up to 4 days. ● Aprepitant is used on day 1, continue at 80 mg po daily on days 2-3. ● Dexamethasone continued on days 2 through 4 at 8 mg PO/IV. ● 5-HT3 RA e.g palonosetron is only given on day 1 due to its long half life.
Moderate emetic risk: 30-90 % frequency of acute emesis	2 or 3 drugs ● NK1 RA + 5-HT3 RA + dexamethasone ● 5-HT3 RA + dexamethasone ● Palonosetron + olanzapine + dexamethasone	● Dexamethasone, olanzapine or aprepitant. ● Palonosetron only give on day 1.
Low emetic risk: 10-30% frequency of acute emesis	1 drug ● Dexamethasone 8-12 mg PO/IV once ● Metoclopramide 10-20 mg PO/IV once ● Prochlorperazine 10 mg PO/IV once ● 5-HT3 RA : ondansetron 8-16 mg PO once	● Routine prophylaxis beyond the first day is not recommended. ● Breakthrough antiemetic treatment is provided as needed not routinely.
Minimal emetic risk: <10% frequency of acute emesis (majority of mAbs)	No routine prophylaxis.	No routine prophylaxis, antiemetics are reserved for breakthrough symptoms only.

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Breakthrough CINV Management

- Persistent CINV despite prophylactic antiemetic therapy can occur both during acute and delayed phases after chemotherapy.
- Initial assessment should include a thorough review of:
  - emetic risk, disease status, adequacy of current antiemetic regimen.
  - rule out other underlying causes like electrolyte imbalance, gastrointestinal obstruction, brain metastasis.
- The cornerstone of breakthrough CINV management is the addition of an agent from a different drug class preferably on a scheduled dosing regimen rather than PRN.
- For subsequent chemotherapy cycles, consider adding agents not previously used.

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Navari RM. Treatment of Breakthrough and Refractory Chemotherapy-Induced Nausea and Vomiting. *BioMed Res Int.* 2015;2015:595894. doi:10.1155/2015/595894.
Jordan K, de Azambuja E, García del Barrio MÁ, et al. Going beyond the 2023 MASCC and ESMO guideline update for the prevention of chemotherapy- and radiotherapy-induced nausea and vomiting. *Eur J Cancer.* 2025;115451. doi:10.1016/j.ejca.2025.115451.

Breakthrough CINV Management Continued

- Olanzapine (2.5- 10 mg PO daily) is recommended as first-line for breakthrough CINV if not used prophylactically. NK1 receptor antagonists are also effective. May consider other route if unable to tolerate oral meds.
- Other options include dopamine antagonists (metoclopramide, haloperidol, promethazine), lorazepam, dronabinol, additional 5-HT3 receptor antagonists or dexamethasone.
- Refractory cases, multiple agents or alternating routes may be necessary.
- Supportive measures include ensuring adequate hydration, correcting electrolyte abnormalities and considering antacids for dyspepsia.

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Overview of CINV in Oral Chemotherapy

- Prophylaxis and treatment for acute and delayed CINV are guided by the risk for emetogenicity similarly to non oral regimen.
- The National Comprehensive Cancer Network (NCCN) recommends scheduled 5-HT3 receptor antagonist (palonosetron, ondansetron) or olanzapine for moderate to high emetic risk oral chemotherapy before and during therapy.
- For low or minimal emetic risk agents, PRN antiemetics with metoclopramide, prochlorperazine, 5-HT3 RAs are recommended.
- Multi day regimens require coverage for each day of therapy and for 2 days after completion.
- Avoid dexamethasone in patients at risk of steroid related adverse effects.

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Non-Pharmacologic and Supportive Interventions

- Adjuncts to pharmacologic therapy in patients with persistent symptoms or at risk for anticipatory CINV.
- Individualized nutrition education and support from a dietitian to reduce the severity of nausea and CINV.
- Ginger supplementation in conjunction with antiemetics, to reduce nausea with high emetic chemotherapy.
- Behavioral interventions such as desensitization, relaxation, cognitive distraction for anticipatory CINV.
- Music therapy may also reduce anticipatory CINV and relieve the severity of delayed vomiting.
- physical exercise and support system can reduce CINV and improve quality of life.

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Gala D, Wright HH, Zigori B, et al. Dietary Strategies for Chemotherapy-Induced Nausea and Vomiting: A Systematic Review. *Clin Nutr (Edinb)*. 2022;41(10):2147-2155. doi:10.1016/j.clnu.2022.08.003.
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The Role of Pharmacist in Optimizing and Management of CINV

- Pharmacist can reference guidelines to identify emetogenic drugs and ensure appropriate antiemetic regimen based on classified risk.
- Pharmacist can promote the safe and effective use of antiemetic drugs by evaluating potential for drug interactions, contraindications and safety profiles (e.g QT prolongation with 5-HT3 RA)
- Pharmacist can educate other healthcare professionals on how to optimize CINV management (over vs under prophylaxis), improve patient outcomes and reduce healthcare utilization.

Knowledge Check

Review Question #1

Which type of chemotherapy induced nausea and vomiting (CINV) is defined by its onset occurring despite adequate prophylaxis?

A. Anticipatory
B. Acute
C. Breakthrough
D. Refractory
E. Delayed

Review Question #2

Which of the following statements are true when it comes to CINV treatment?

A. CINV caused by highly emetogenic chemotherapy agents should be prevented with a serotonin/dopamine antagonist + Neurokinin-1 receptor antagonist.
B. Dexamethasone side effect includes sedation, immunosuppression and psychosis.
C. Olanzapine should only be given intravenously.
D. Granisetron may prolong QTc in certain patients populations
E. Fosaprepitant works by inhibiting binding of substance P and should be given as a one time dose of 150 mg in adults.

Review Question #3

Which of the following statements are true when it comes to CINV treatment?

A. Breakthrough CINV should be treated with the same agents used for primary prophylaxis.
B. Limit dose of IV ondansetron to 16 mg in patients at risk of QT prolongation.
C. Palonosetron has the lowest risk of QT prolongation
D. Dexamethasone is safe to use in patients with systemic fungal infections.

References

1. National Comprehensive Cancer Network. NCCN Clinical Practice Guidelines in Oncology (NCCN Guidelines®) for Antiemesis. Version 2.2025. Published May 12, 2025. Accessed November 2, 2025. https://www.nccn.org/professionals/physician_gls/pdf/antiemesis.pdf
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Questions?